LUC@BJTU Weihai Campus

Fundamentals of Communication Systems (CNSCC160) Tutorial 3

Multiple Choice Question

1. In analogue-to-analogue conversion, why is modulation necessary for transmitting signals like audio frequencies?

- A) To prevent signal mixing
- B) To reduce transmission cost
- C) To adjust antenna height for efficient radiation
- D) To increase signal power

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2. Which of the following types of mechanism allows for the accommodation of multiple users on the same communication link?

- A) Amplitude Modulation (AM)
- B) Frequency Modulation (FM)
- C) Multiplexing
- D) Phase Modulation (PM)

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3. In amplitude modulation (AM), what primarily defines the envelope of the AM waveform?

- A) The carrier frequency
- B) The amplitude of the modulating signal
- C) The frequency of the modulating signal
- D) The phase shift of the modulating signal

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4. Which of the following is NOT an advantage of modulation in communication systems?

- A) Improved system reliability
- B) Increased transmission range
- C) Reduced need for bandwidth
- D) Efficient radiation in radio systems

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5. In Frequency Modulation (FM), what aspect of the carrier signal is modified in response to the amplitude of the modulating signal?

- A) Carrier phase
- B) Carrier frequency
- C) Carrier amplitude
- D) Carrier power

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6. What is the relationship between the bandwidth of an AM signal and the modulating signal?

- A) AM bandwidth is equal to the modulating signal's bandwidth.

- B) AM bandwidth is double the bandwidth of the modulating signal.
- C) AM bandwidth is half the bandwidth of the modulating signal.
- D) AM bandwidth depends on the carrier frequency, not the modulating signal.
  Answer: B

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7. In a simple AM demodulation circuit, which component is primarily responsible for recovering the modulating signal from the carrier?

- A) Resistor
- B) Capacitor
- C) Diode
- D) Inductor

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8. What is the primary drawback of amplitude modulation (AM) compared to other modulation techniques like FM or PM?

- A) AM requires higher transmission power.
- B) AM is highly susceptible to noise.
- C) AM uses a more complex demodulation process.
- D) AM requires a larger bandwidth than FM.

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9. In Frequency Shift Keying (FSK), what remains constant while the frequency is varied to represent the digital data?

- A) Phase
- B) Carrier frequency
- C) Amplitude
- D) Modulation index

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10. Which of the following digital modulation schemes combines both amplitude and phase modulation to achieve higher efficiency?

- A) Amplitude Shift Keying (ASK)
- B) Phase Shift Keying (PSK)
- C) Quadrature Amplitude Modulation (QAM)
- D) Frequency Shift Keying (FSK)

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11. The Shannon Capacity theorem suggests that the maximum rate of data transfer for error-free communication is influenced by which two factors?

- A) Signal power and transmission distance
- B) Bandwidth and noise levels
- C) Modulation technique and power
- D) Carrier frequency and data rate

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12. In Binary Phase Shift Keying (BPSK), how many signal elements are used, and how are they differentiated?

- A) Two signal elements, differentiated by frequency
- B) Two signal elements, differentiated by amplitude
- C) Two signal elements, differentiated by phase
- D) Three signal elements, differentiated by phase

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13. In an FM signal, what happens to the bandwidth as the modulation index (β) increases?

- A) The bandwidth decreases.
- B) The bandwidth remains constant.
- C) The bandwidth increases.
- D) The bandwidth fluctuates based on the carrier frequency.

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14. Which of the following correctly describes the concept of symbol rate in digital modulation?

- A) It is the same as the data rate in bps.
- B) It refers to the number of bits transmitted per second.
- C) It is the number of signal elements transmitted per second.
- D) It is unrelated to the modulation technique used.

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15. What modulation technique is typically used to transmit digital data over optical fibre?

- A) Amplitude Shift Keying (ASK)
- B) Frequency Shift Keying (FSK)
- C) Quadrature Amplitude Modulation (QAM)
- D) On-Off Keying (OOK)

Understanding

1. Distinguish between baseband transmission and broadband transmission.

2. We send a voice signal from a microphone to a recorder. Is this baseband or broadband transmission?

3. Can we say if a signal is periodic or nonperiodic by just looking at its frequency-domain plot? How?

4. Which of the following defines the theoretical maximum bit rate of a noiseless channel?
    - Nyquist theorem
    - Shannon capacity

5. Distinguish between a signal element and a data element.

6. Which of the four digital-to-analogue conversion techniques (ASK, FSK, PSK, or QAM) is the most susceptible to noise?

7. What is the difference between AM (Amplitude Modulation) and FM (Frequency Modulation)?

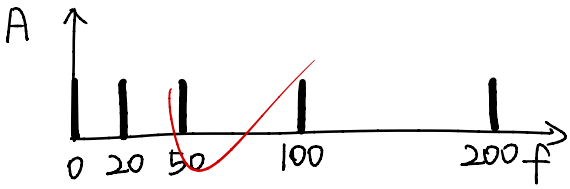
8. What is the difference between Analogue and Digital Modulation? Use Amplitude modulation (AM) and Amplitude shift Keying (ASK) as an example in your explanation.

9. Which characteristic is changed to represent the digital data in each of the following digital modulations (digital-to-analogue conversions)?
    - ASK
    - FSK
    - PSK

10. Which characteristic is changed to represent the analogue signal in each of the following analogue modulations (analogue-to-analogue conversions)?
    - AM
    - PM
    - FM

Understanding

1. What is the bandwidth of a signal that can be decomposed into five sine waves with frequencies of 0, 20, 50, 100, and 200 Hz? All peak amplitudes are the same. Draw the bandwidth.



Bit rate = $\frac{1}{\text{Bit interval}}$

2. Give the bit rate for each of the following signals.
   a. A signal in which 1 bit lasts 0.001 s — a: 1000 bps
   b. A signal in which 1 bit lasts 2 ms — b: $\frac{1}{2}\times10^{3}$ = 500 bps
   c. A signal in which 10 bits last 20 µs — c: $\frac{10}{20\times10^{-6}}$ = 5×10^{5} bps

3. A device is sending out data at the rate of 1000 bps.
   a. How long does it take to send out 10 bits? $\frac{10}{1000} = 10^{-2}$ s
   b. How long does it take to send out a single character (8 bits)? $\frac{8}{1000} = 8\times10^{-3}$ s
   c. How long does it take to send a file of 100,000 characters? $\frac{100{,}000\times8}{1000} = 800$ s

4. A line has a signal-to-noise ratio of 1000 and a bandwidth of 4000 kHz. What is the maximum data rate supported by this line?

   $C = 4000\times10^{3}\cdot\log_2(1+1000) = 3.98\times10^{7}$ bps.

5. A signal with 200-mW power passes through 10 devices, each with an average noise of 2 µW. What is the SNR? What is the SNRdB? 20 µW

   $SNR = \frac{200\times10^{-3}}{20\times10^{-6}} = 10^{4}$ SNRdB = 40 dB

6. We need to upgrade a channel to a higher bandwidth.
   a. How is the rate improved if we double the bandwidth? Double
   b. How is the rate improved if we double the SNR? little influence.

7. Assume that a voice channel occupies a bandwidth of 4 kHz. We need to multiplex 10 voice channels with guard bands of 500 Hz using FDM. Calculate the required bandwidth.

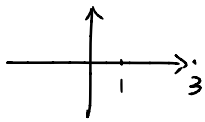
   $4000\times10 + 500\times9 = 44500$ Hz

8. Which signal has a wider bandwidth, a sine wave with a frequency of 100 Hz or a sine wave with a frequency of 200 Hz? We don't know.

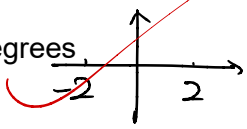
9. When implementing Amplitude Modulation, what is the modulated signal's bandwidth of a 4 kHz speech signal if transmitted on a carrier of 30 kHz?

   4kHz + 30kHz = 34kHz

10. Draw the constellation diagram for the following:
    a. BASK, two symbols having peak amplitude values of 1 and 3



    b. BPSK, two symbols having a peak amplitude value of 2 but 180 degrees difference between their phases?



11. Calculate the symbol rate / baud rate for the given bit rate and type of modulation.
    a. 2000 bps, BFSK r=1 S = 2000 bps
    b. 4000 bps, BASK r=1 S = 4000 bps

12. Baseband refers to the original frequency range of a signal before it is converted, or modulated, to a different frequency range. For example, an audio signal may have a

baseband range from 20 to 20,000 Hz. Some simple antennas usually have a length close to half the wavelength of the transmitted signal. What is the length of the antenna if an audio signal having a frequency of 20 KHz is directly transmitted without modulation? Note: Suppose the propagation speed is 3×10^8 m/s.

$$X = \frac{3\times10^8}{20\times10^3} = 15000\,m$$

$$= 15\,km$$